

KesslerX

From Collision Warning to Accountable Action.

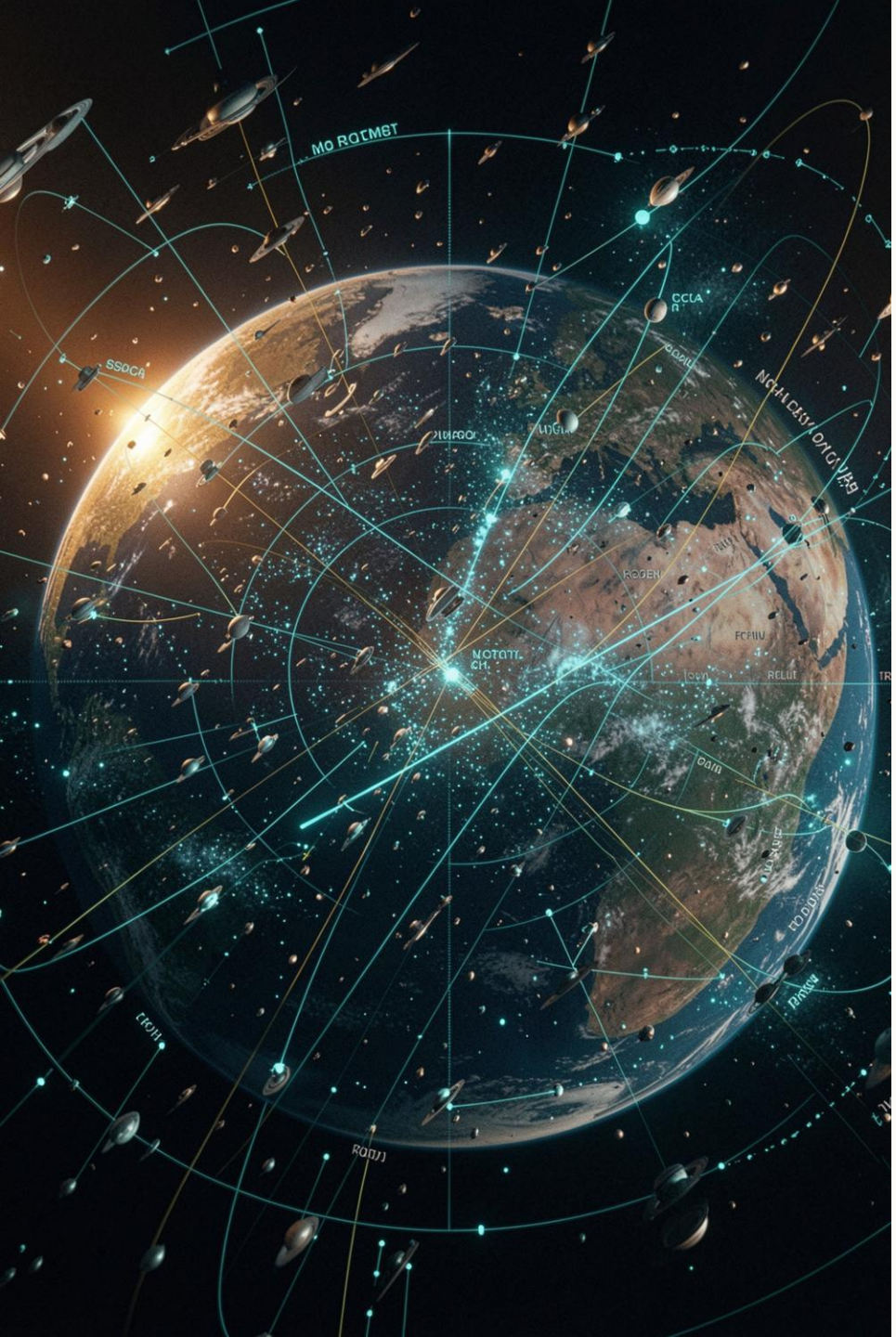
Orbital Traffic Management & Collision-Avoidance Intelligence

PROTONEDGE ANISH MOGAM — LEAD MADHURA LAKADE SHREYAS GORE

FAR AWAY HACKATHON

KESSLER-X.VERCEL.APP

SEE THE ENVIRONMENT. UNDERSTAND THE RISK. COORDINATE RESPONSIBLY.



Space Is Becoming a Traffic-Management Problem

High-impact context

40,000+ tracked objects • 1.2M+ debris objects >1 cm (published estimates)

The operational gap

Detecting a close approach is only the beginning. Operators need provenance, timeliness, ownership, and a clear next action.

Fragmented workflow

CATALOG → MAP → SPECIALIST ANALYSIS → MESSAGING → MANUAL HANDOFF → REPORT (disconnected tools impede timely, accountable response)

Orbital risk is not only a prediction problem. It is a coordination problem.

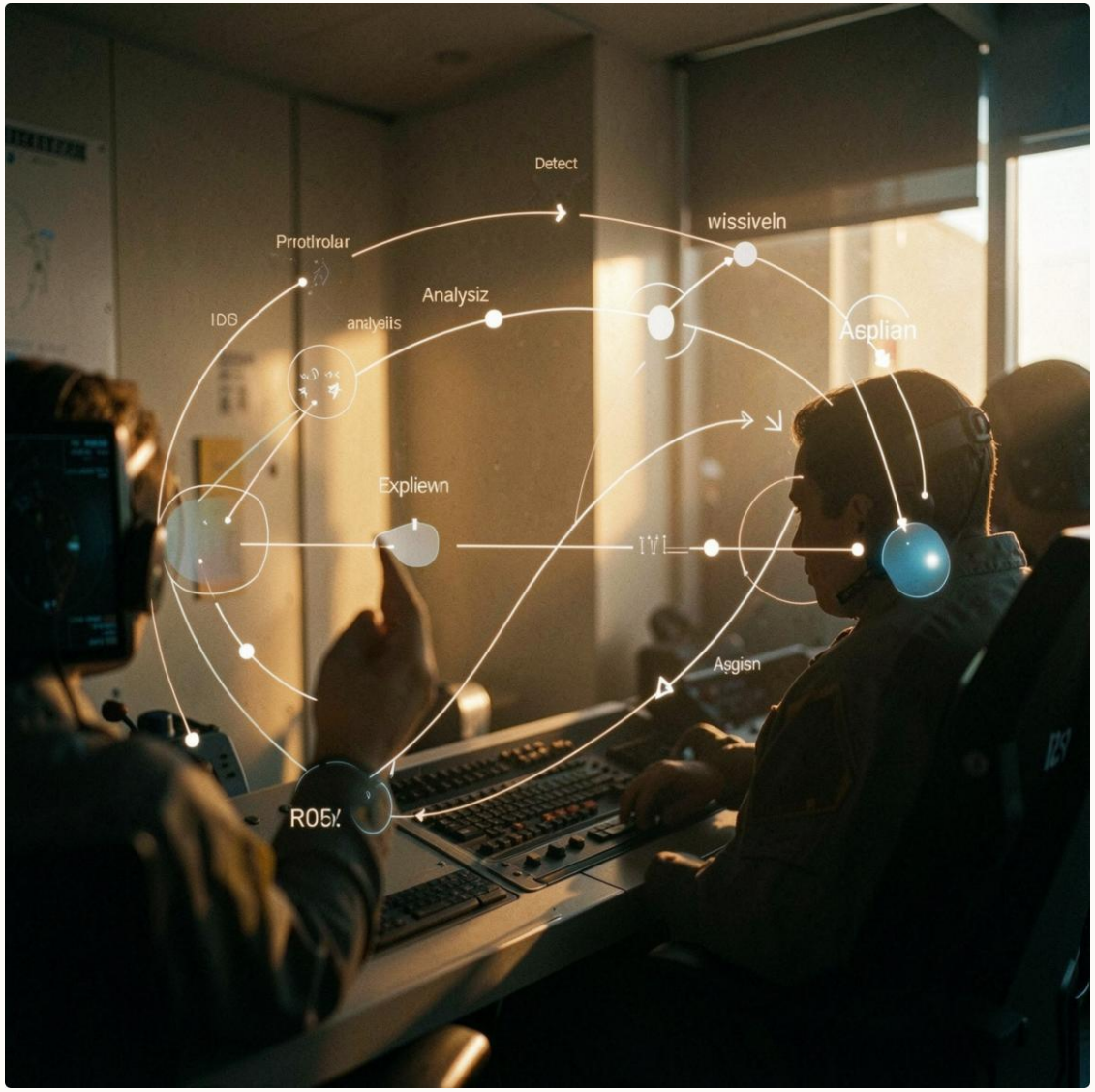
A Warning Is Not an Operational Decision

Traditional response (fragmented)



- Detect → open separate tools
- Rebuild context manually
- Message other operators, then assign

KesslerX (connected)



Observe → Detect → Prioritize → Analyze → Explain → Assign → Review

ONE EVENT → ONE WORKSPACE

Meet KesslerX



3D ORBITAL AWARENESS

Cesium-based visualization, satellite+debris layering, search & filter, orbital context.



COLLISION INTELLIGENCE

Event prioritization, TCA, miss distance, relative velocity, risk indicators.



CONJUNCTION ANALYSIS

TLE inputs → propagation → state vectors → advisory probability charts.



MISSION CONTEXT

Telemetry, ground-station data, space weather and mission feed integrated.



OPERATIONAL WORKFLOW

Event workspace, owner assignment, next action, deadline, handoff history.



AI COPILOT

Summaries, evidence discovery, handoff drafting—operator-in-the-loop assistance.

SEE THE ENVIRONMENT → UNDERSTAND THE RISK → COORDINATE RESPONSIBLY

From Red Alert → Accountable Case

1

01 DETECT

Potential conjunction identified (CX-9921-A). Initial advisory generated from propagation.

2

02 PRIORITIZE

TCA, miss distance, probability → priority score for operator attention.

3

03 ANALYZE

TLE ingestion, propagation comparisons, relative motion charts.

4

04 UNDERSTAND

Telemetry, space weather, mission context, evidence aggregation.

5

05 ASSIGN

Owner identified, next action assigned, deadline set—single accountable case.

6

06 HANDOFF

Contextual handoff note, evidence links, timeline, and verification checklist.

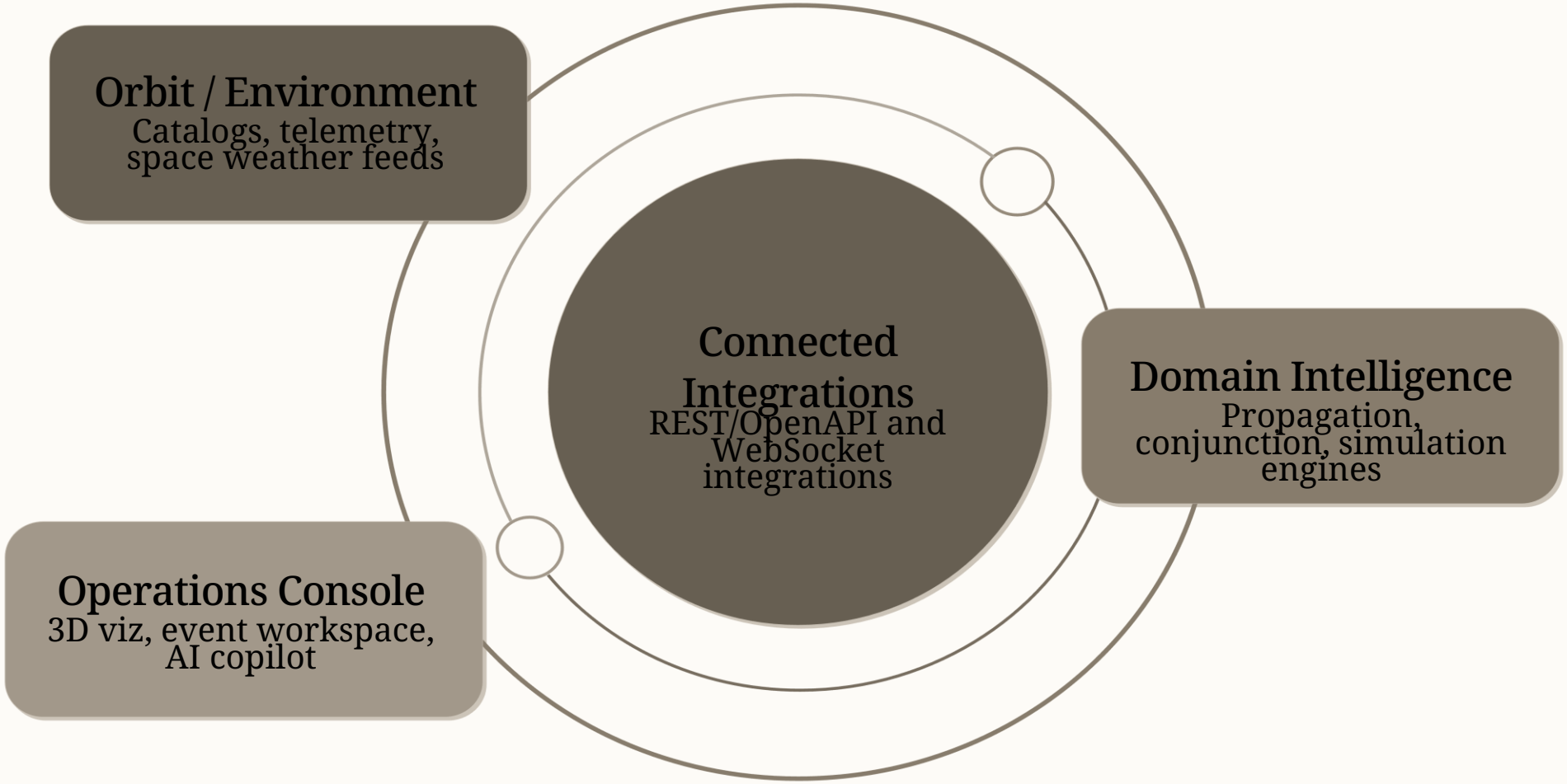
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07 REVIEW

Decision state, report generation, and historical record for audit.

Alert → Event → Owner → Next Action. LIVE DEMO: kessler-x.vercel.app

Built as a Connected System — Not a Mockup



Experience
React web console — route-level operational modules and event-driven UI.

Visualization
Cesium 3D environment, layered satellites, debris, timelines and filters.

Domain Intelligence
Satellite catalog, TLE-defined objects, SGP4-style propagation, conjunction engine, simulation.

Operational State
Collision events, alerts, event workspace, ownership, handoffs, reports.

Context
Telemetry, ground stations, space weather, external feeds.

Integration
REST / OpenAPI, WebSocket telemetry, external orbital & weather sources.

Technology: React · Cesium · REST/OpenAPI · TLE / SGP4-style propagation · WebSocket · AI · Simulation

Prototype exposes calculations, event state and evidence rather than hiding them behind a single opaque risk score.

Not Another Satellite Tracker.

WHERE ARE THE OBJECTS?

1

WHAT IS THE RISK?

Conjunction analysis + trend & uncertainty.

2

WHO OWNS IT?

Owner, action, deadline—accountability recorded.

3

WHAT CHANGED?

Versioned evidence, data freshness and delta detection.

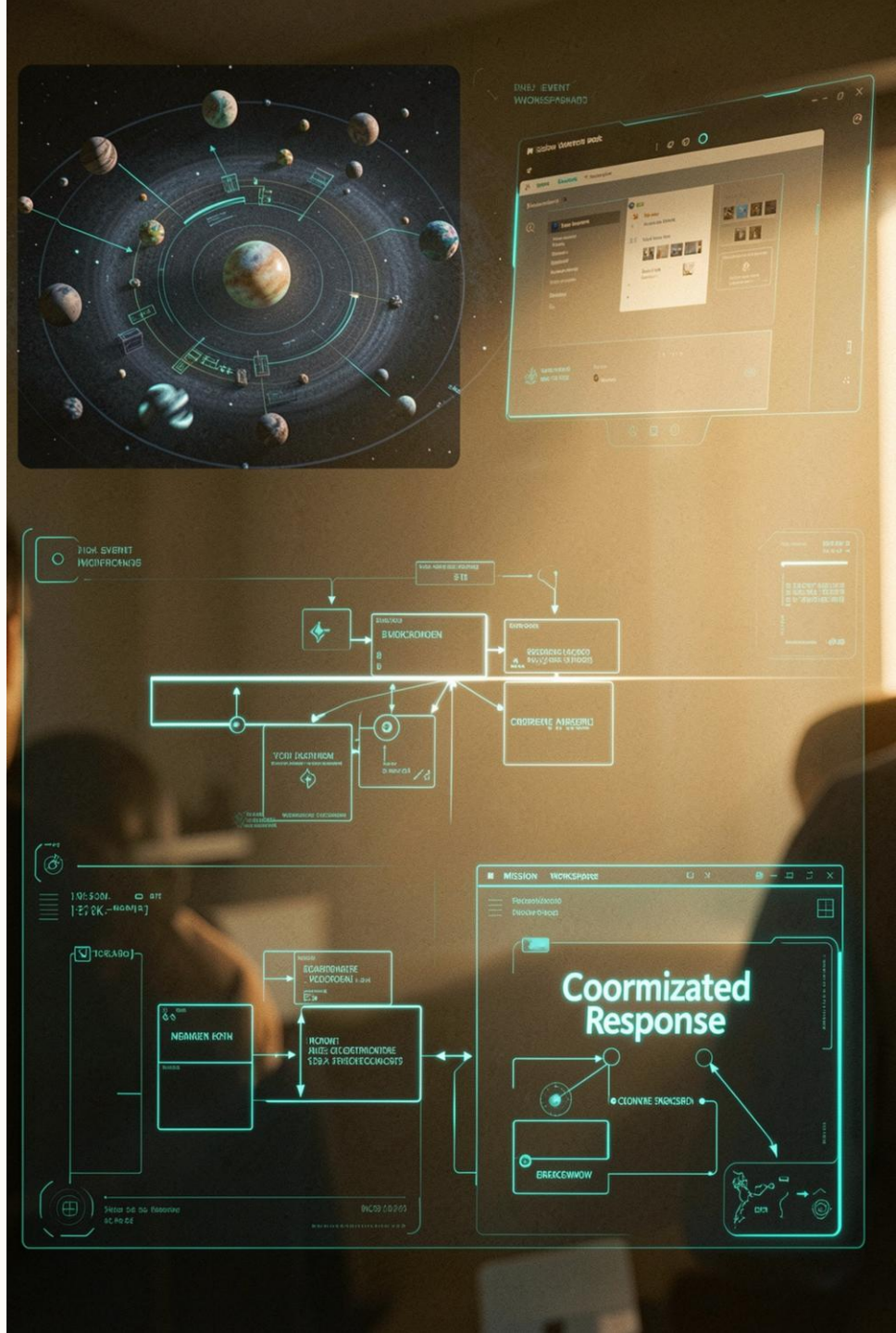
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WHAT HAPPENS NEXT?

Assigned next action, handoff, and review workflow.

3D AWARENESS + CONJUNCTION ANALYSIS + ENVIRONMENTAL CONTEXT + AI ASSISTANCE + OWNERSHIP + HANDOFFS + REPORTING = KESSLERX

EVENT-CENTERED WORKFLOW — the coordination layer between detection and accountable human action.



AI That Assists. Humans That Decide.

AI CAN ASSIST

- Summarize an event and its evidence
- Explain risk metrics and uncertainty
- Locate relevant telemetry and space-weather context
- Draft handoff notes and checklists for operators

AI CANNOT DECIDE

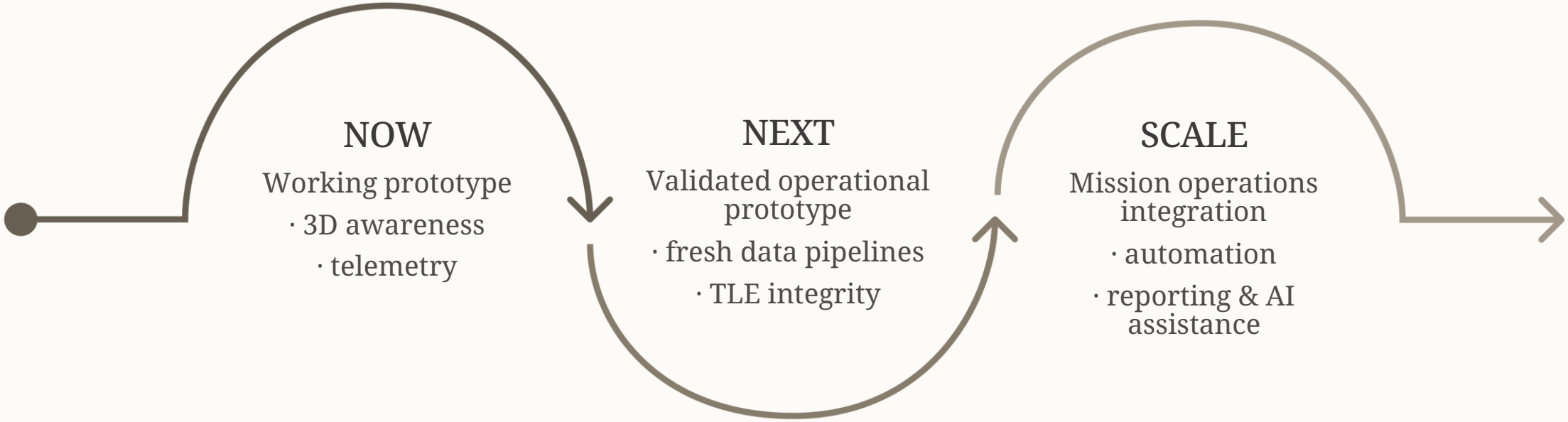
- Execute spacecraft maneuvers
- Replace mission-control approval
- Hide uncertainty or invent evidence
- Claim flight-certified guidance

HUMAN-IN-THE-LOOP

Evidence → Source → Timestamp → Model / Version → Uncertainty → Human Review → Approval

Responsible AI is an engineering requirement: evidence-grounded outputs, explicit uncertainty, and auditable model metadata.

Trust Before Automation



NOW — Working Prototype

3D awareness, conjunction workflow, event workspace, telemetry, space weather, simulation, AI assistance, reporting.

NEXT — Validated Operational Prototype

Fresh data pipelines, TLE freshness & integrity validation, trusted SGP4 test vectors, evidence-grounded AI, server-side authorization, audit trails.

SCALE — Mission Operations Integration

Device & network security, monitoring, approval workflows, operational integrations and deployment pipelines.

OUR ROADMAP IS NOT: MORE AUTOMATION. IT IS: MORE TRUST.

KesslerX

FROM COLLISION WARNING TO ACCOUNTABLE ACTION.

01 SEE

3D orbital environment — satellites, debris and trajectories.

02 UNDERSTAND

Risk metrics, propagation-based analysis, telemetry and space-weather context.

03 COORDINATE

Owner assignment, handoffs, timelines and auditable reports.

PROTONEDGE ANISH MOGAM — LEADER MADHURA LAKADE SHREYAS GORE FAR AWAY HACKATHON

kessler-x.vercel.app — LIVE DEMO

See the environment. Understand the risk. Coordinate responsibly.